

Hanguk Telephony

Self-Built Call Routing & Call Recording — Implementation Plan

Build our own phone system in-house: no Mediateka, no cloud PBX subscription, no monthly payment to any telephony company. One small box is bought once; everything else is free software running on a spare computer.

Decision

- One-time hardware purchase — OK
- A spare PC / cheap mini-PC will run the software
- Start with a simple “Press 1, Press 2” menu (no AI voice agent, no monthly AI phone bill)
- **One SIM card is enough for now** — buy the 1-port gateway, not the 8/16-port version

Prepared for: Hanguk (Tashkent) — owner review copy

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Companion research: docs/research/call-routing-ivr.md & docs/research/mobile-call-capture.md

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1. The Big Picture

Right now, one phone number can only ring one phone. If four people call your business number at the same time, only one gets through and the other three hear a busy tone. To fix this without paying a phone company every month, we put a small box between the phone line and our staff, and run free software that decides who a call goes to.

Someone calls your business SIM number

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GSM Gateway box (GoIP-1) — turns the SIM's call into a computer signal

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Asterisk / FreePBX software on a spare PC (free, runs 24/7)

- Plays the menu: “Press 1 for admissions, 2 for payments...”
- Puts the call in a fair queue and rings free staff in turn
- Skips whoever is marked “away” in the CRM

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Staff answer on a free softphone app on their computer or phone

What you pay for, ever: the gateway box (once), a PC (once, or free if you already have a spare one), and your normal SIM card bill (which you already pay). Nothing recurring goes to any PBX or CRM-telephony company — Asterisk is free, open-source software.

The honest trade-off: since nobody else is running this for you, you become the “IT department” for it. Someone sets it up once and keeps the PC turned on. That is the real cost of “no third party” — time, not money.

2. Hardware to Buy

You only need **one** piece of new hardware to start: a 1-SIM GSM-to-VoIP gateway. It is a small box, about the size of a home Wi-Fi router, with one SIM card slot and one antenna.

The gadget to buy

GoIP-1

What it is: a 1-port GSM/VoIP gateway. You put your business SIM card inside it; it connects to your PC over Wi-Fi/Ethernet using the standard SIP protocol, so Asterisk can answer and make calls through that SIM.

Why this one: it is the cheapest way to prove the whole system works before spending more. You said one SIM is enough — this is the exact model sized for that.

What to say to a seller: “I need a 1-port GSM VoIP gateway, brand GoIP (model GoIP-1), or an equivalent single-SIM SIP/VoIP GSM gateway that works with Asterisk/FreePBX.” Any shop that sells VoIP/PBX or CCTV/security equipment for offices will recognize this category, even if they don't stock the GoIP brand specifically — a Dinstar or Yeastar 1-port gateway is an acceptable substitute.

Typical price: roughly \$50–90 (marketplace listings; confirm with a local reseller, shipping/customs to Uzbekistan will add to this).

Everything else you need (most of it you probably already have)

Item	Do you need to buy it?	Notes
GoIP-1 gateway (1 SIM)	Yes — the one purchase	See above.
A PC to run the software	Only if you don't have a spare one	Any old desktop/laptop works, or a cheap mini-PC (~\$50–100), or a Raspberry Pi 4. It must stay switched on.
Business SIM card	You already have this	The same number your business currently uses.
Headset or softphone app	Free	A free app (e.g. Zoiper, Linphone) on each staff computer/phone acts as their “extension.” No special desk phones required to start.
Ethernet cable / Wi-Fi	You already have this	The gateway just needs to join your office network.

Do not buy (for now)

- **GoIP-8 / GoIP-16 / Dinstar 4–32 port gateways** (\$350–\$1,140+) — these hold multiple SIM cards for larger call centres. Skip them; one SIM is enough today.

- **Any Mediateka-style subscription or cloud PBX** (UzCloud VPBX, Zadarma, Sipuni) — these charge a monthly fee and are exactly what we are avoiding.
- **Desk IP phones** — not needed yet; a free softphone app does the same job while the team is small.

3. Where to Buy It in Tashkent

No single confirmed Tashkent stock listing for the GoIP-1 specifically could be verified from here — treat the options below as starting points to call, not guaranteed in-stock links.

- **AliExpress / international marketplaces** — search “GoIP-1 GSM VoIP gateway”. Reliable stock and price transparency, but allow 2–4 weeks shipping plus customs into Uzbekistan.
- **Local VoIP/PBX and telecom equipment resellers in Tashkent** (e.g. Eurolux and similar IT/telecom supply shops) — call and ask specifically: “Do you sell or can you order a 1-port GSM-to-SIP VoIP gateway, such as GoIP-1, Dinstar, or Yeastar TG100?” Confirm the exact model, price in UZS, and whether it ships with a power adapter and antenna suited to local GSM bands.
- **Ask any office-PBX / CCTV installer** — companies that install office phone systems or security camera networks in Tashkent often already import this exact category of device and can source one even if it isn't on their shelf.

What to ask any seller, word for word: “I need a single-SIM GSM-to-VoIP gateway that supports the SIP protocol and works with Asterisk or FreePBX. One SIM port is enough. Can you get me a GoIP-1, or an equivalent from Dinstar or Yeastar?”

4. Software Setup (free, one-time)

Software	What it does	Cost
Asterisk	The free “engine” that answers calls, plays menus, and manages queues.	Free
FreePBX	A free web dashboard on top of Asterisk, so you click to set things up instead of editing text files.	Free
RasPBX / Incredible PBX / Docker image	Ready-made install packages so this is a few-hour setup, not a from-scratch build.	Free

Runs comfortably on a spare PC, a cheap mini-PC, or a Raspberry Pi 4 (Pi 5 has some reported quirks — test before buying one specifically for this). Ongoing cost is just electricity for the machine — no license fee, ever.

5. The Phone Menu (IVR)

This is the “Press 1 for...” menu callers hear. It is built in FreePBX by recording (or uploading) short voice prompts and mapping each digit to a destination. No AI, no ongoing cost.

Caller dials in

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Short recorded greeting plays (Uzbek + Russian)

“1” → Admissions queue

“2” → Payments queue

“3” → Existing student support

“0” or no answer → front desk / owner

Record the prompts once — a phone voice memo is good enough quality — upload them in FreePBX, done.

Future option, not part of this build: letting callers just speak instead of pressing digits needs an AI voice agent, and every mainstream platform for that (Vapi, Retell, Bland, Twilio) is itself a paid third-party service — skipped for now since it conflicts with the no-third-party decision.

6. Call Queue — Sharing Calls Fairly

Asterisk's built-in queue system is the same technology every commercial call centre uses. It comes free with the software above.

Strategy	What it does	Best for
Ring All	Rings every free staff member at once; first to answer gets it.	Simplest, good for 2–4 people.
Round Robin	Cycles through staff in order, remembering where it left off.	Simple, guaranteed fairness.
Least Recent	Rings whoever has gone longest without a call.	Fairest option once you have more than a couple of people — usual default.
Fixed Order	Always tries the same order (e.g. front desk, then backup, then owner).	Reception → escalation.

Skip whoever's away: the CRM already tracks each staff member as online/away/busy/offline. A small bridge script can log staff in or out of the phone queue automatically whenever their CRM status changes, so marking yourself “away” in the CRM also stops your phone from ringing.

Hold music / “you are caller number 3” / unlimited wait time: all standard and free, built in — nothing extra to build. The wait queue can be left completely open-ended (timeless), with a periodic reassurance message instead of a hang-up timer.

7. Build Order — Step by Step

- 1 **Buy one GoIP-1** and plug your business SIM into it. Set up Asterisk/FreePBX on the spare PC. Confirm you can receive and make a call through it.
- 2 **Add the phone menu (IVR)** in front of a single test queue.
- 3 **Add the queue** with the “Least Recent” or “Ring All” strategy and 2+ staff extensions — a free softphone app on each computer/phone is enough to start.
- 4 **Bridge staff presence (optional):** a small script that reads the CRM's staff status and logs staff in/out of the phone queue to match.
- 5 **Scale up later, only if needed:** move to a 4-port or 8-port gateway once the single line proves itself and you know how many simultaneous callers you actually get.
- 6 **Log calls into the CRM:** Asterisk can send a message on every call event, feeding the same student/lead matching system already used for other calls, so a routed call attaches to the right student automatically.

8. Recording Calls

Routing (who a call rings) and recording (capturing the audio) are two separate problems solved with two different pieces of hardware/software. Here is the plan for recording, reusing what you already own.

8.1 Your existing YINGXIN desk phone

You already own a SIM-card desk phone with a hardware REC button. This device sidesteps every Android restriction discussed below because the SIM plugs directly into it — the phone itself has full access to the call audio, no app or operating system in the way.

- **Automatic recording:** this feature is standard across the product line. Press the physical **MENU** button and look for a recording-settings option (Off / Manual / Auto-all-calls) — switch it to Auto if it isn't already.
- **Getting recordings out:** check for a removable SD/TF memory card slot — common on this product line, up to 32GB. If present, just pop the card out and read it on any computer, no special software needed.
- **Bringing it into the CRM:** a small “Import recordings” screen (not yet built) lets staff upload a batch of files, match each one to the caller and student, and it flows into the same transcription/analysis pipeline as every other call. This is a small, fast piece of work — no foreground service, no Android permission fights.
- **Limitation:** only covers calls taken at that one desk, on that one SIM.

8.2 Staff personal mobile phones (“Call Sync”) — longer-term

For calls staff take away from the desk on their own phones, a native module in the existing staff app can log every call (number, direction, duration) automatically and attach it to the right student — this part works reliably on any Android phone. Two-way audio recording depends on the phone brand:

Phase	What it delivers	Effort
Phase 0 — Probe	Test build confirms which recording method works on the actual staff phone before committing.	2–3 days
Phase 1 — Metadata sync	Every call (number, duration, direction, SIM) logged and linked to the student automatically, even before audio works.	5–8 days
Phase 2 — Audio	Recording turned on per phone brand: Xiaomi/Redmi phones usually have this built in already; Samsung and others need a fallback method with mixed reliability.	7–10 days
Phase 3 — Hardening	Battery-saver survival, consent notice, admin dashboard, roll out to more staff.	4–6 days

Note on the current staff phone (Samsung Galaxy A33 5G): Samsung's own built-in recording is off by default in Uzbekistan's region, so the fallback method (an app called Cube ACR) is the realistic Phase 0 starting point — expect it to need testing, and possibly one-sided audio on some calls.

Legal note (unverified): Uzbek law is not clearly settled on whether one side of a call may record without saying so. Get a local lawyer's opinion before rolling this out beyond the owner's own phone, and add a short recording notice to the student agreement regardless — standard call-centre practice.

9. Cost Summary

Item	One-time cost	Monthly cost
GoIP-1 gateway (1 SIM)	~\$50–90 + shipping/customs	\$0
PC / mini-PC to run Asterisk	\$0 if reusing a spare machine, else ~\$50–100	\$0 (just electricity)
Asterisk + FreePBX software	\$0 (free, open-source)	\$0
Softphone apps for staff	\$0 (free apps)	\$0
SIM card	Already owned	Whatever you already pay your carrier
YINGXIN desk phone recording	\$0 (already owned)	\$0
Mobile Call Sync module	Developer time only (~3–4 weeks total)	\$0

Bottom line: one hardware purchase under \$100, plus a spare PC you may already own, replaces every recurring telephony subscription.

10. Risks & Honest Trade-offs

Risk	What it means	How it's handled
You are now the IT department	No vendor to call when something breaks.	Software is mature and very widely documented; start with one SIM to learn the system before scaling.
Shipping / customs for the gateway	Cost and delivery time not fully confirmed for Uzbekistan.	Ask local resellers first; budget extra time and a customs fee if ordering internationally.
Mobile recording is device-dependent	Not every staff phone can record both sides of a call.	Phase 0 tests the actual phone first; the desk-phone and metadata-only paths still work regardless.
Legal clarity on one-party recording	Uzbek law is not clearly settled.	Get a lawyer's opinion before wide rollout; add a recording notice either way.

11. Sources

Full research with every link is kept in the project repository:

- [docs/research/call-routing-ivr.md](#) — call queuing (ACD) and IVR research, gateway pricing
- [docs/research/mobile-call-capture.md](#) — Android call-recording feasibility and the YINGXIN desk phone

Key external sources referenced in that research:

- Asterisk queues.conf.sample — github.com/asterisk/asterisk
- FreePBX Queues module guide — sangomakb.atlassian.net
- GoIP/Dinstar gateway pricing — goantifraud.com, buymyvoip.com
- GoIP + Asterisk setup walkthroughs — ixnfo.com, blog.telarvostore.com
- Self-hosted PBX comparison 2026 — bigiron.cc
- Raspberry Pi telephone exchange guide — magazine.raspberrypi.com
- Android call-recording platform rules — developer.android.com
- Samsung One UI native recording region list — notebookcheck.net

Hanguk — internal planning document. Not for external distribution.